RUNGTA COLLEGE OF ENGINEERING & TECHNOLOGY
BHILAI CHHATTISGARH

AIML TECHNICAL TRAINING
MINOR PROJECT REPORT

PRODUCT REVIEW SENTIMENT ANALYSIS USING MACHINE LEARNING

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Semester	5th Semester
Academic session	2026–27

Project Details:-

Item	Details
Project title	Product Review Sentiment Analysis
Category	AIML Technical Training — Group Minor Project
Programme	B.Tech 5th Semester
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Technical trainer	Mr. Shekhar Kumar
Academic session	2026–27
Source code	https://github.com/kundangupta04/product-review-sentiment
Live demo	https://kundangupta04-product-review-sentiment-app-hcrrae.streamlit.app/

Declaration

We declare that this report describes the work carried out by our group for the AIML Technical Training minor project titled "**Product Review Sentiment Analysis** Using TF-IDF and Machine Learning." The source code is maintained in a public repository by group member Kundan Gupta. All external datasets, libraries and documentation used in the project are acknowledged in the References. The results were produced by running the project code on the described dataset.

We understand that the report, implementation, screenshots and viva explanation must be consistent with the work actually performed.

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1. INTRODUCTION

1.1 Background

Customer reviews provide useful information about product quality, value and customer experience. However, manually analysing large numbers of reviews is difficult. Sentiment analysis is a Natural Language Processing (NLP) task that estimates whether text expresses a positive or negative opinion.

1.2 Problem Statement

To develop a simple web-based machine-learning application that accepts product review text and predicts whether the review expresses positive or negative sentiment.

1.3 Objectives

- Prepare a labelled dataset for binary sentiment classification.
- Clean review text and preserve meaningful negation words such as “not”.
- Convert text into numerical features using TF-IDF.
- Train a Logistic Regression classifier and evaluate it using suitable metrics.
- Provide a Streamlit interface for user-entered reviews.

1.4 Scope and Significance

The system is an educational prototype for English-language review classification. It demonstrates how data preparation, feature engineering, supervised learning and a simple user interface can be combined in an AIML application. It is not a replacement for human judgement and may perform poorly on sarcasm, ambiguous reviews or product categories different from its training data.

1.5 Existing and Proposed Approach

Manual reading is slow at scale, while rule-based systems may struggle with context. The proposed approach uses TF-IDF with Logistic Regression because it is lightweight, practical for sparse text data and suitable for a student-level machine-learning project.

2. DATASET AND METHODOLOGY

2.1 Dataset

The reference repository uses the Amazon Fine Food Reviews dataset, with review text and star rating as the main fields. Ratings 4–5 are labelled Positive; ratings 1–2 are labelled Negative; neutral 3-star reviews are excluded. The repository describes selecting a 20,000-review sample and then removing missing, empty and duplicate records.

The reference README reports 19,289 reviews after cleaning: 16,309 positive and 2,980 negative. These counts belong to the reference implementation and must be verified if the group uses a different dataset file or cleaning process.

2.2 Text Preprocessing

The documented preprocessing includes lowercasing, removing HTML tags and URLs, expanding selected contractions, removing punctuation and extra whitespace, and handling missing or empty text. Negation words should be preserved because expressions such as “not good” have a different meaning from “good”.

2.3 TF-IDF Feature Extraction

TF-IDF converts review text into numerical features. It weights terms according to their frequency in a review and how common they are across the dataset. The reference configuration uses up to 5,000 features and unigrams plus bigrams. The vectoriser should be fitted only on training data to prevent test-set leakage.

2.4 Logistic Regression

Logistic Regression is a supervised classification algorithm that learns feature weights to distinguish positive from negative reviews. The reference repository specifies max_iter=1000, random_state=42 and balanced class weights. The final report should reflect the exact settings used by the group's code.

2.5 Evaluation Metrics

Metric	Purpose
Accuracy	Overall proportion of correct predictions.
Precision	How many predictions for a class are correct.
Recall	How many actual examples of a class are identified.
F1-score	Balances precision and recall.
Confusion matrix	Shows true and false predictions for both classes.

2.6 Functional Requirements

- Load and validate the review dataset.
- Derive positive/negative labels from star ratings and exclude neutral ratings.
- Clean review text and remove missing, empty and duplicate records.
- Split data into training and testing subsets before fitting TF-IDF.
- Train a Logistic Regression classifier and save the fitted pipeline.
- Display a sentiment prediction and, if implemented, a confidence score for user-entered text.
- Handle empty input and missing model files without an uncontrolled crash.

2.7 Non-Functional Requirements

- Usability: the interface should be simple for a first-time user.
- Performance: a single prediction should complete promptly on ordinary hardware.
- Maintainability: data preparation, training, prediction and interface code should be separated where practical.
- Reproducibility: record dataset version, parameters, random seed and software versions.
- Reliability: the same preprocessing and TF-IDF pipeline must be used at training and prediction time.

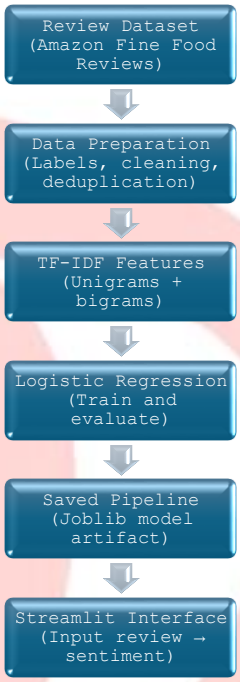
2.8 Software and Hardware Requirements

Component	Reference Requirement/Role
Hardware	Laptop or desktop; no dedicated GPU is required for this classical ML pipeline.
Operating System	Windows,Linux or macOS
Python	Python 3.10 or newer, as specified by the reference README.
Pandas/NumPy	Tabular data handling and numerical operations.
Scikit-learn	TF-IDF, Logistic Regression, pipeline and evaluation metrics.
Matplotlib	Exploratory charts and confusion-matrix visualisation.
Joblib	Persistence of the fitted model pipeline.
Streamlit	Browser-based user interface.
GitHub	Version control and source hosting

3. SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Architecture

The architecture below summarises the documented reference workflow from data preparation through model deployment.



3.2 User Workflow

1. The user opens the Streamlit application.
2. The user enters a product review in the input field.
3. The application validates that the input is not empty.
4. The saved pipeline applies the same text transformation and TF-IDF representation used during training.
5. Logistic Regression predicts Positive or Negative sentiment.
6. The interface displays the predicted class and a confidence score if supported by the deployed version.

3.3 Implementation and Project Structure

The public repository separates code into source modules for data loading, preprocessing, dataset cleaning, exploratory data analysis, feature generation, training, evaluation, prediction and tests. The top-level app.py provides the Streamlit interface; the fitted pipeline is stored under models/ and reports/figures are stored under outputs/.

File / folder	Purpose
src/data_loader.py	Load raw data, create labels, inspect
src/preprocessing.py	clean_text() function
src/clean_dataset.py	Apply cleaning; drop missing, empty and duplicate rows
src/eda.py	Exploratory analysis and saved charts
src/features.py	Stratified split and TF-IDF pipeline
src/train.py	Train Logistic Regression and save the pipeline
src/evaluate.py	Metrics and confusion matrix
src/predict.py	load_model() and predict_sentiment()
src/compare_models.py	Optional Multinomial Naive Bayes comparison
src/test_*.py	Unit, app and end-to-end tests
app.py	Streamlit web interface
check_setup.py	Verifies libraries and folders
models/, outputs/	Saved pipeline; figures and reports

3.4 Ethical and Responsible Use

Review text can contain personal or order-related details, so users should not paste sensitive information into a public demo. Predictions are statistical estimates and may be wrong for sarcastic, ambiguous, mixed or out-of-domain reviews. The system should not be the sole basis for decisions affecting customers or sellers.

4. RESULTS AND TESTING

4.1 Testing

Testing is performed to evaluate the functionality and performance of the Product Review Sentiment Analysis system. The system uses TF-IDF for text feature extraction and Logistic Regression to classify reviews into Positive and Negative categories.

The following test cases are used to check the system's behaviour under different input conditions.

Test Case	Expected Behaviour
Positive review	Predict Positive sentiment
Negative review	Predict Negative sentiment
Mixed review	Return a sentiment prediction
Empty input	Display a validation message
Very long review	Process the text without errors

These tests help verify the prediction functionality, input handling, and reliability of the application. The test cases should be executed using the group's own application before the final report is submitted.

4.2 Model Evaluation and Results

The reference implementation evaluates the model on a test dataset containing 3,858 reviews, including 3,262 Positive and 596 Negative reviews.

Table 4.1: Reference model performance

Evaluation Metric	Positive	Negative
Precision	96.95%	64.77%
Recall	91.63%	84.23%
F1-score	94.22%	73.23%

The reference model achieves an overall accuracy of **90.49%** and a macro F1-score of **0.8372**. Precision indicates how many predictions for a class are correct, while recall measures how many actual instances of that class are identified. The F1-score balances precision and recall.

4.3 Confusion Matrix

A confusion matrix compares the actual sentiment labels with the model's predictions.

Table 4.2: Reference confusion matrix

Actual Class	Predicted Negative	Predicted Positive
Negative	502	94
Positive	273	2,989

The model correctly classifies 502 Negative reviews and 2,989 Positive reviews. However, 94 Negative reviews are incorrectly classified as Positive, while 273 Positive reviews are incorrectly classified as Negative.

These results show that the model performs well on Positive reviews but needs improvement in Negative-class precision. Since the dataset is imbalanced, accuracy alone is insufficient to evaluate its performance.

4.4 Screenshots of Results

The following screenshots should be included as evidence of the application's functionality and evaluation.

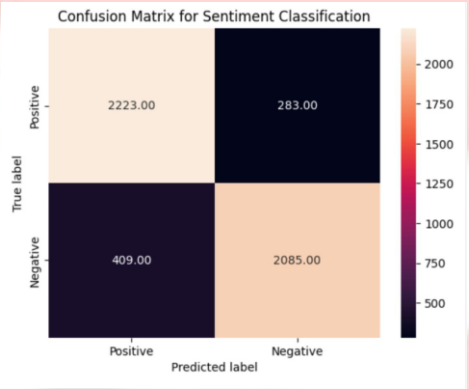


Figure 4.1: Confusion Matrix

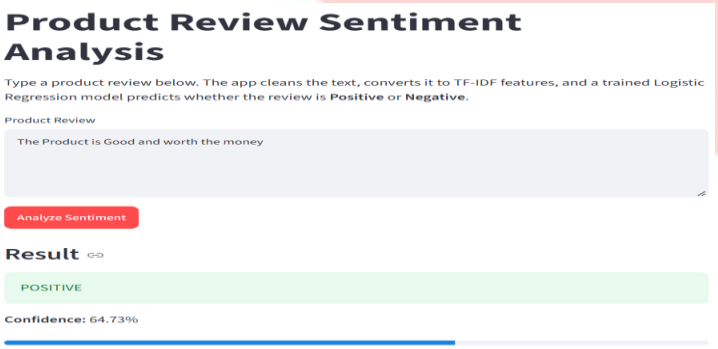


Figure 4.2: Screenshot showing the prediction for a Positive review.

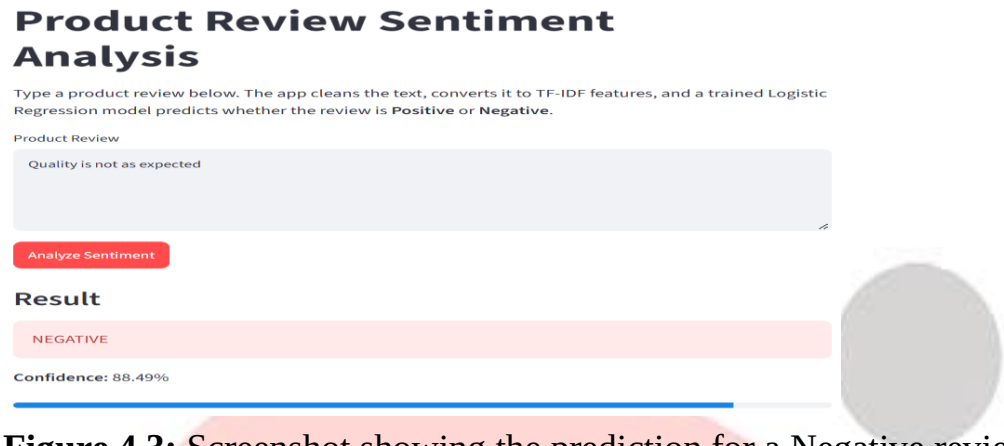


Figure 4.3: Screenshot showing the prediction for a Negative review.

5. CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Product Review Sentiment Analysis project demonstrates the use of Natural Language Processing (NLP) and machine learning to classify customer reviews as Positive or Negative. The system combines text preprocessing, TF-IDF feature extraction, Logistic Regression, and a Streamlit web interface to provide sentiment predictions.

The reference implementation reports an accuracy of 90.49%, demonstrating the potential of this approach for automated review analysis. However, the difference in performance between Positive and Negative reviews highlights the need for further improvement. The project provides practical experience in data preparation, model development, performance evaluation, and application deployment.

5.2 Limitations

The current system has the following limitations:

- It supports only Positive and Negative sentiment categories.
- Imbalanced data can affect classification performance.
- TF-IDF and Logistic Regression have limited understanding of sarcasm and complex context.
- A model trained on food reviews may not generalise well to other product categories.
- Prediction confidence does not guarantee that the output is correct.

5.3 Future Scope

The project can be improved through the following extensions:

1. Neutral sentiment classification: Add a Neutral category to handle reviews that are neither clearly Positive nor Negative.
2. Larger datasets: Train and evaluate the model on reviews from different product categories.
3. Model comparison: Compare Logistic Regression with other machine learning algorithms and tune their parameters.
4. Improved predictions: Apply threshold tuning and confidence calibration to improve classification reliability.
5. Sentiment dashboard: Develop visualisations showing sentiment distribution and common customer complaints.
6. Advanced NLP: Explore transformer-based models for better contextual understanding when computational resources permit.

These improvements could make the system more reliable and useful for analysing customer feedback.

6. REFERENCES

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